THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	High resistance draws almost no current.
2	**(c)** 6 ohm	R2 = rho 2L / 4A = R / 2.
3	**(b)** V against I	Ohm's law.
4	**(c)** A is true but R is false.	A is true; R is false: copper has low resistivity.
5	**(b)** 9 R	L is tripled, A is one third.
6	**(d)** 5 ohm	R = V / I.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Greater collisions give more resistance.
8	**(b)** R / n^2	Each part R/n; parallel gives R/n^2.
9	**(a)** Both A and R are true and R is the correct explanation of A.	A low-resistance meter in series does not change the current it is measuring.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true ($R = \rho l/A$ and ρ changes with temperature), but R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	I = 4.8×10^{-3} A (1); n = 4.8×10^{-3} / 1.6×10^{-19} = 3×10^{16} (1).	1+1
12	R = 220^2 / 800 = 60.5 ohm (1); H = P t = 800 x 300 = 2.4×10^5 J (1).	1+1
13	R = V^2 / P (1); Ratio R100 : R25 = 1 : 4 (1).	1+1
14	Two halves of 2 ohm each in parallel (1); R = 1 ohm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Atoms vibrate more, increasing collisions with electrons (1); Resistance increases (1); Current decreases (I = V / R) (1).	1+1+1
16	P = 300 W (1); I = 300 / 220 = 1.36 A (1); R = 220 / 1.36 = 161.3 ohm (1).	1+1+1
17	Ammeter in series should not add to the resistance (1); Voltmeter in parallel should not draw current (1); Else the readings are altered (1).	1+1+1
	OR: Area ratio = 1 : 4 (1); R1 : R2 = 4 : 1 (1); Parallel: I1 : I2 = 1 : 4 (1).	1+1+1
18	Voltmeter has a very high resistance (1); Total resistance becomes very high (1); Almost no current flows and the bulb will not glow (1).	1+1+1
	OR: R = 4 / 0.2 = 20 ohm (1); I = 10 / 20 = 0.5 A (1); P = 5 W (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $R = V/I = 2/0.4 = 5\ \Omega$ (1). (ii) A straight line passing through the origin (1). (iii) At constant temperature the current through a conductor is directly proportional to the potential difference across its ends (1). (iv) Each half has resistance $2.5\ \Omega$ (1); in parallel $R = 2.5/2 = 1.25\ \Omega$ (1).	1+1+1+2
20	New R = 9 x 10 = 90 ohm (1); Each third = 30 ohm (1); 1/R = 3 / 30 (1); R = 10 ohm (1); Resistance comes back to the original (1).	1+1+1+1+1
	OR: R60 = 806.7 ohm; R40 = 1210 ohm (1); Total = 2016.7 ohm; I = 0.109 A (1); V60 = 88 V; V40 = 132 V (1); P60 = 9.6 W (1); P40 = 14.4 W (1).	1+1+1+1+1

Mistakes to avoid:

- Adding the reciprocals incorrectly: for resistors in parallel $\frac{1}{R_p} = \sum \frac{1}{R}$ and the final answer is R_p, not the sum of reciprocals.
- Connecting an ammeter in parallel or a voltmeter in series; the ammeter goes in series and the voltmeter in parallel.

- Thinking that resistance is directly proportional to cross-sectional area; $R \propto \frac{1}{A}$ and $R \propto l$.
- Using $H = I^2Rt$ with t in minutes or hours; time must be in seconds (and energy in kWh needs power in kW and time in hours).
- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 27 W	P = V^2 (1/2 + 1/4) = 36 x 0.75.
2	**(b)** 4 ohm	Each branch 8 ohm; two in parallel gives 4 ohm.
3	**(a)** Both A and R are true and R is the correct explanation of A.	The fuse protects appliances.
4	**(d)** 3 ohm and 6 ohm	3 + 6 = 9 and 18 / 9 = 2.
5	**(b)** 2 Q	R halves; H = V^2 t / R doubles.
6	**(d)** A is false but R is true.	A is false; R is true: parallel resistance is less than each.
7	**(a)** 60 W	Same current, larger R dissipates more power.
8	**(b)** 144 W	Series: V^2/(2R) = 36; parallel: V^2 (2/R) = 4 times.
9	**(c)** A is true but R is false.	In series the current is the same but the potential difference divides; R is false.
10	**(d)** A is false but R is true.	A fuse wire has a low melting point so that it melts easily; A is false and R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	P is proportional to V^2 (1); P = 100 x (1/4) = 25 W (1).	1+1
12	P = 2.2 kW (1); E = 4.4 kWh, cost = Rs 26.4 (1).	1+1
13	Current divides inversely to resistance (1); I = 4 x 12 / 16 = 3 A (1).	1+1
14	Cold filament has low resistance so a large current flows (1); Sudden heating weakens it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	I = 12 / 6 = 2 A (1); P1 = 4 W, P2 = 8 W, P3 = 12 W (1); Total = 24 W = V I (1).	1+1+1
16	Joule is a very small unit (1); 1 kWh = 3.6×10^6 J (1); kWh is convenient (1).	1+1+1
17	Current is lower for the same power (1); Heat I^2 R t in the lines is reduced (1); Less power is wasted (1).	1+1+1
	OR: Stretched resistance = 4R (1); Each half = 2R (1); Parallel: R (1).	1+1+1
18	Resistance decreases as length falls (1); P = V^2 / R increases (1); The coil gets hotter (1).	1+1+1
	OR: V = 2 x 6 = 12 V (1); I in 12 ohm = 1 A (1); Total = 3 A (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2+3+6=11\ \Omega$ (1). (ii) $\frac{1}{R}=\frac{1}{2}+\frac{1}{3}+\frac{1}{6}=1$, so $R=1\ \Omega$ (1). (iii) $I=V/R=6/1=6$ A (1). (iv) $6/2=3$ A, $6/3=2$ A, $6/6=1$ A (1); $3+2+1=6$ A which equals the total current (1).	1+1+1+2
20	R1 = 2 / 0.4 = 5 ohm (1); R2 = 6 / 1.2 = 5 ohm (1); R is the same, so ohmic (1); P = V I = 6 x 1.2 (1); P = 7.2 W (1).	1+1+1+1+1
	OR: V / I = 10 ohm each time (1); So Ohm's law holds (1); P = 3 x 0.3 = 0.9 W (1); rho = R A / L = 10×10^{-6} / 5 (1); rho = 2×10^{-6} ohm m (1).	1+1+1+1+1

Mistakes to avoid:

- Adding the reciprocals incorrectly: for resistors in parallel $\frac{1}{R_p} = \sum \frac{1}{R}$ and the final answer is R_p, not the sum of reciprocals.
- Connecting an ammeter in parallel or a voltmeter in series; the ammeter goes in series and the voltmeter in parallel.
- Thinking that resistance is directly proportional to cross-sectional area; $R \propto \frac{1}{A}$ and $R \propto l$.
- Using $H = I^2Rt$ with t in minutes or hours; time must be in seconds (and energy in kWh needs power in kW and time in hours).
- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 800 J	H = I^2 R t; 16 / 4 x 1 / 2 x 400.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Currents are added algebraically (Kirchhoff's rule is not required), so current is a scalar.
3	**(c)** 275 W	P is proportional to V^2.
4	**(d)** 1 kWh	0.25 x 4.
5	**(a)** Both A and R are true and R is the correct explanation of A.	The field moves with a high speed along the wire.
6	**(b)** 5 : 3	R is proportional to 1/P.
7	**(c)** Reducing the temperature	Lower temperature decreases R.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R = V^2 / P.
9	**(a)** Both A and R are true and R is the correct explanation of A.	R states why parallel connection is preferred.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; tungsten is chosen for its very high melting point, which R does not mention.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heat produced depends on I^2 R t (1); The fuse melts when the current exceeds the limit (1).	1+1
12	R1 + R2 = 25 and R1 R2 = 100 (1); R1 = 5 ohm and R2 = 20 ohm (1).	1+1
13	R = 806.7 ohm (1); I = 110 / 806.7 = 0.136 A (1).	1+1
14	H = I^2 R t (1); H becomes 8 times (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	0.5 (4 + r) = 0.25 (8 + r) (1); 2 + 0.5 r = 2 + 0.25 r (1); r = 0, emf = 2 V (1).	1+1+1
16	Each side 3 ohm (1); One side 3 ohm in parallel with the other three sides 9 ohm (1); R = 27 / 12 = 2.25 ohm (1).	1+1+1
17	Same current flows (1); Nichrome has higher resistance (1); H = I^2 R t is greater for nichrome (1).	1+1+1
	OR: I = 220 / 55 = 4 A (1); P = 880 W (1); E = 0.88 x 5 = 4.4 kWh, cost Rs 22 (1).	1+1+1
18	Total resistance doubles (1); Current halves (1); Each bulb glows dimmer (1).	1+1+1
	OR: Q = 2 x 30 = 60 C (1); V = 20 V (1); E = V Q = 1200 J (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $1.5 \times 2 = 3$ kWh (1). (ii) $4 \times 0.1 \times 5 = 2$ kWh (1). (iii) 3.6×10^6 J (1). (iv) Total per day $= 5$ kWh (1); for 30 days $=$ 150 units, bill $= 150 \times 6 =$ Rs 900 (1).	1+1+1+2
20	(a) All in series: 3 + 6 + 9 = 18 ohm (1); (b) All in parallel: 1/R = 1/3 + 1/6 + 1/9 = 11/18 (1); R = 1.64 ohm (1); (c) 3 ohm and 6 ohm in parallel give 2 ohm (1); Adding 9 ohm in series gives 9 + 2 = 11 ohm (1).	1+1+1+1+1
	OR: At 220 V: I = 22000 / 220 = 100 A (1); Loss = I^2 R = 10000 x 2 = 20000 W (1); At 2200 V: I = 10 A (1); Loss = 100 x 2 = 200 W (1); High-voltage transmission reduces the loss 100 times (1).	1+1+1+1+1

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- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
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